your own projects. Each chapter includes exercises to test your progress, and sample code and answers are available online at https://github.com/cltrudeau/django-in-action.

This book is divided into three parts. The first part covers Django's core components, the second part delves into Django's built-in tools, and the third part covers how to use third-party libraries to build different kinds of sites.

In part 1, you start to build RiffMates. You will write views and register them against URLs. You'll learn about the template engine and how to load and render templates in your views to output HTML to the browser. You will also learn how to interact with the database doing queries on musician and venue data, while using the Django Admin to manage it all.

Part 2 shows you the power of Django and all the nifty stuff built into it. You'll turn RiffMates into a multiuser site and learn how to customize the provided views dealing with logins and password management. Once you've got user accounts happening, you will add the ability to get input from your users, including uploaded files.

Part 3 focuses on how key third-party libraries can help you extend the functionality of your website. Django was originally designed as a content management system framework, but its community has created all sorts of pluggable tools to do other things. You can build websites with REST APIs; deal with data import and export; and interact with popular JavaScript frameworks, like HTMX.

Happy web coding!

Summary

- Django maps a URL to a view, which is either a function or a class that returns a document to the browser.
- A Django view can render a document by composing pieces of HTML through a template engine.
- The model—view—controller architectural pattern divides the parts of a program into three pieces. The model contains data and business logic, the view is for visualizing the data, and the controller is for interacting with it.
- An object relational mapping is a way of abstracting database tables with programming classes.
- Django includes tools for managing multiuser websites.
- Single-page applications move the view and controller parts of the MVC into the browser.
- Django can be used as the backend for a single-page application.